Isometry Pursuit: Convex Recovery of Orthonormal Column Submatrices

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Abstract

Isometry pursuit is a convex algorithm for identifying orthonormal column-submatrices of wide matrices. It consists of a vector normalization followed by multitask basis pursuit. Applied to Jacobians of putative coordinate functions, it helps identity locally isometric embeddings from within interpretable dictionaries. We provide theoretical and experimental results justifying this method, including a proof with realistic assumptions that such isometric submatrices, should they exist, are contained within the obtained support. For problems involving coordinate selection and diversification, it offers a synergistic alternative to greedy and brute force search.

1 Background

Our algorithm is motivated by a classic question in numerical linear algebra.

1.1 Problem

Our goal is, given a matrix $X \in \mathbb{R}^{D\times P}$, to select a subset $S \subset [P]$ with $|S| = D$ such that $X_{.S}$ is as orthonormal as possible in a computationally efficient way. To this end, we define a ground truth loss function that measures orthonormalness, and then introduce a surrogate loss function that convexifies the problem so that it may be efficiently solved. Figure **??** and **??** provide a visual depiction of this problem when $D = 2$ and $P = 30$.

1.2 Isometry

Definition 1 *The **differential** of a smooth map $\phi : \mathcal{M} \to \mathcal{N}$ between D dimensional manifolds $\mathcal{M} \subseteq \mathbb{R}^B$ and $\mathcal{N} \subseteq \mathbb{R}^P$ is a map in tangent bases $x_1 \dots x_D$ of $T_\xi\mathcal{M}$ and $y_1 \dots y_D$ of $T_{\phi(\xi)}\mathcal{N}$ consisting of entries*

$$D\phi(\xi) = \begin{bmatrix} \frac{\partial\phi_1}{\partial x_1}(\xi) & \dots & \frac{\partial\phi_1}{\partial x_D}(\xi) \\ \vdots & & \vdots \\ \frac{\partial\phi_D}{\partial x_1}(\xi) & \dots & \frac{\partial\phi_D}{\partial x_D}(\xi) \end{bmatrix}. \tag{1}$$

It is not always necessary to explicitly estimate tangent spaces when applying this definition. The most commonly encountered manifolds are vector spaces for which the tangent spaces are trivial. This is the case for full-rank tabular data, for which isometry has a natural interpretation as a type of diversification, and often for the latent spaces of deep learning models. In this case, $B = D$.

Definition 2 *A map ϕ between D dimensional submanifolds with inherited Euclidean metric $\mathcal{M} \subseteq R^B$ and $\mathcal{N} \subseteq R^P$ ϕ is an **isometry at a point** $\xi \in \mathcal{M}$ if*

$$D\phi(\xi)^T D\phi(\xi) = I_D. \tag{2}$$

That is, ϕ is an isometry at ξ if $D\phi(\xi)$ is orthonormal.

The applications of pointwise isometry are themselves manifold. Pointwise isometric embeddings faithfully preserve high-dimensional geometry. For example, Local Tangent Space Alignment (**?**), Multidimensional Scaling (**?**) and Isomap (**?**) non-parametrically estimate embeddings that are as isometric as possible. Another approach stitches together pointwise isometries selected from a dictionary to form global embeddings (Kohli et al., 2021). The method is particularly relevant since it constructs such isometries through greedy search, with putative dictionary features added one at a time.

That $D\phi$ is orthonormal has several equivalent formulations. The one motivating our ground truth loss function comes from spectral analysis.

Proposition 1 *The singular values $\sigma_1 \dots \sigma_D$ are equal to 1 if and only if $U \in \mathbb{R}^{D \times D}$ is orthonormal.*

On the other hand, the formulation that motivates our convex approach is that orthonormal matrices consist of D coordinate features whose gradients are orthogonal and of unit length.

Proposition 2 *The component vectors $u_1 \dots u_D \in \mathbb{R}^B$ form a orthonormal matrix if and only if, for all $d_1, d_2 \in [D], \langle u_{d_1}, u_{d_2} \rangle = \begin{cases} 1 \; d_1 = d_2 \\ 0 \; d_1 \neq d_2 \end{cases}$.*

1.3 Matrix functions suitable for measuring isometry

A function of a rectangular matrix is suitable for measuring isometry provided it is The *condition number* (Golub and Wilkerson https://gdz.sub.uni-goettingen.de/id/PPN362160546_0009?*tify* = ,*measurestheratioofthelargestandsmallestsingularvectorsofamatrix*A.*Thishasbeenexamplebeenusedtoassessput*

A class of functions suitable for measuring isometry are S-divergences (**?**). S-divergences define a metric $D(P, P')$ on positive definite matrices P, P'. If those matrices are $X^T X$ and I_D, the S-divergence distance is

$$D(X^T X, I_D) = (\log\det(\frac{X^T X + I_D}{2}) - \frac{1}{2}\log\det(X^T X))^{\frac{1}{2}}. \tag{3}$$

This is a log singular value penalty. This is because, first of all,

$$\frac{1}{2}\log\det(X^T X) = \frac{1}{2}\log\prod_{d=1}^{D}\sigma_d(X)^2 \tag{4}$$

$$= \log\prod_{d=1}^{D}\sigma_d(X) \tag{5}$$

$$= \sum_{d=1}^{D}\log\sigma_d(X). \tag{6}$$

Secondly, I_D and $X^T X$ are diagonalizable in the same basis, so

$$\log\det(\frac{X^T X + I_D}{2}) = \log\prod_{d=1}^{D}\frac{\sigma_d(X)^2 + 1}{2} \tag{7}$$

$$= \sum_{d=1}^{D}\log\frac{\sigma_d(X)^2 + 1}{2}. \tag{8}$$

Then,

$$D(X^TX, I_D)^2 = \sum_{d=1}^{D}(\log \frac{\sigma_d(X)^2+1}{2} - \log \sigma_d(X)) \tag{9}$$

$$= \sum_{d=1}^{D} \log \frac{\sigma_d(X)^2+1}{2\sigma_d(X)} \tag{10}$$

$$= \sum_{d=1}^{D} \log \frac{\sigma_d(X)+\sigma_d(X)^{-1}}{2}. \tag{11}$$

TODO: explain how works for PSD not just PD.

1.4 Interpretability and isometry

Our motivating example is the selection of data representations from within sets of putative coordinates: the columns of a provided wide matrix. Compared with Sparse PCA (**????**), we seek a low-dimensional representation from the set of these column vectors rather than their span.

This method applies to interpretability, for which parsimony is at a premium. Interpretability arises through comparison of data with what is known to be important in the domain of the problem. This knowledge often takes the form of a functional dictionary. Evaluation of independence of dictionary features arises in numerous scenarios (**?**Koelle et al., 2022; **?**). The requirement that dictionary features be full rank has been called functional independence (Koelle et al., 2022) or feature decomposability (**?**), with connection between dictionary rank and independence via the implicit function theorem. Besides independence, the metric properties of such dictionary elements are of natural interest. This is formalized through the notion of differential.

1.5 Best subset selection

Given a matrix $X \in \mathbb{R}^{D\times P}$, we compare algorithmic paradigms for solving problems of the form

$$\arg \min_{S\in\binom{[P]}{D}} \ell(X_{\cdot S}) \tag{12}$$

where $\binom{[P]}{D} = \{A \subseteq [P] : |A| = D\}$. Brute force algorithms consider all possible solutions. These algorithms are conceptually simple, but have the often prohibitive time complexity $O(C_\ell P^D)$ where C_ℓ is the cost of evaluating ℓ. Greedy algorithms consist of iteratively adding one element at a time to S. This algorithms have time complexity $O(C_\ell PD)$ and so are computationally more efficient than brute force algorithms, but can get stuck in local minima. Formal definitions are given in Section 2.1.

Sometimes, it is possible to introduce an objective which convexifies problems of the above form (**???**). Solutions

$$\arg\min f(\beta) : Y = X\beta \tag{13}$$

to the overcomplete regression problem $Y = X\beta$ are a classic example (**?**). When $f(\beta) = \|\beta\|_0$, this problem is non-convex, and is thus suitable for greedy or brute algorithms, but when $f(\beta) = \|\beta\|_1$, the problem is convex, and may be solved efficiently via interior-point methods. When the equality constraint is relaxed, Lagrangian duality may be used to reformulate as a so-called Lasso problem, which leads to an even richer set of optimization algorithms (**?**).

The form of basis pursuit that we apply is inspired by the group basis pursuit (Koelle et al., 2022; **?**). Multitask basis pursuit is a special form of group basis pursuit which we call differently since grouping is dependent only on the structure of matrix-valued response variable y. The objective $f(\beta) = \|\beta\|_{1,2} := \sum_{p=1}^{P} \|\beta_{p\cdot}\|_2$ creates joint sparsity across entire rows $\beta_{p\cdot}$ (**???**). It was used in Koelle et al. (2022) to select independent subsets among interpretable features.

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2 Supplement

This section contains algorithms, proofs, and experiments in support of the main text.

2.1 Algorithms

We give definitions of the brute and greedy algorithms for the combinatorial problem studied in this paper. The brute force algorithm is computationally intractable for all but the smallest problems, but always finds the global minima.

BruteSearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f)

1: **for** each combination $S \subseteq \{1, 2, \ldots, P\}$ with $|S| = D$ **do**
2: Evaluate $f(X_{.S})$
3: **end for**
4: **Output** the combination S^* that minimizes $f(X_{.S})$

Greedy algorithms are computationally expedient but can get stuck in local optima (Cormen et al., 2009; Russell and Norvig, 2009), even with randomized restarts (Dick et al., 2014).

GreedySearch(Matrix $X \in \mathbb{R}^{D \times P}$, objective f, selected set $S = \emptyset$, current size $d = 0$)

1: **if** $d = D$ **then**
2: **Return** S
3: **else**
4: **Initialize** $S_{\text{best}} = S$
5: **Initialize** $f_{\text{best}} = \infty$
6: **for** each $p \in \{1, 2, \ldots, P\} \setminus S$ **do**
7: **Evaluate** $f(X_{.(S \cup \{p\})})$
8: **if** $f(X_{.(S \cup \{p\})}) < f_{\text{best}}$ **then**
9: **Update** $S_{\text{best}} = S \cup \{p\}$
10: **Update** $f_{\text{best}} = f(\mathcal{X}_{.(S \cup \{p\})})$
11: **end if**
12: **end for**
13: **Return** GreedySearch(X, f, S_{best}, $d + 1$)
14: **end if**

2.2 Proofs

2.2.1 Proof of Proposition ??

In this proof we first show that the penalty $\|\beta\|_{1,2}$ is unchanged by unitary transformation of β.

Proposition 3 *Let $U \in \mathbb{R}^{D \times D}$ be unitary. Then $\|\beta\|_{1,2} = \|\beta U\|$.*

Proof:

$$\|\beta U\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p.} U\| \tag{14}$$

$$= \sum_{p=1}^{P} \|\beta_{p.}\| \tag{15}$$

$$= \|\beta\|_{1,2} \tag{16}$$

□

We then show that this implies that the resultant loss is unchanged by unitary transformation of X.

Proposition 4 *Let $U \in \mathbb{R}^{D\times D}$ be unitary. Then $\widehat{\beta}(UX) = \widehat{\beta}(X)U$.*

Proof:

$$\widehat{\beta}(UX) = \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = UX\beta \tag{17}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ U^{-1}U = U^{-1}UX\beta U \tag{18}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta U \tag{19}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta U\|_{1,2} \ : \ I_D = X\beta U \tag{20}$$

$$= \arg\min_{\beta\in\mathbb{R}^{P\times D}} \|\beta\|_{1,2} \ : \ I_D = X\beta. \tag{21}$$

□

2.2.2 Proof of Proposition ??

In this proof we first show that the orthonormal submatrix $X_{.S}$ is contained within the set $\widehat{\beta}_{MBP}(w(X), I_D)$.

Proposition 5 *Let $X_{.S}$ be a rank D orthonormal submatrix of X. Then $S \in S(\beta) : \beta \in \widehat{\beta}_{MBP}(w(X), I_D))$.*

Proof: By Proposition **??** without loss of generality, let $X_{.S} = I_D$. Also, without loss of generality let $S = \{1 \dots D\}$. We show that $\beta = [I_D \ 0]$ satisfies the KKT conditions for $X = [I_D \ X_{-S}]$.

Recall that the KKT conditions for a convex optimization problem are a set of conditions on an element of the domain of the optimization algorithm that, if satisfied, certify that the element is a solution. To write out the KKT conditions, we define the Lagrangian

$$\mathcal{L}(X, \beta, \nu) = \|\beta\|_{1,2} + \nu^T(I_D - w(X)\beta) \tag{22}$$

where $\nu \in \mathbb{R}^{D\times D}$ is the dual variable.

The KKT conditions are then

- Primal feasibility: $w(X)\beta = I_D$
- Stationarity: there exists a dual variable ν such that $0 \in \partial\|\beta\|_{1,2} - w(X)^T\nu$ where ∂ is the subdifferential operator.

where dual feasibility and complementary slackness are ignorable by virtue of the absence of inequality constraints.

First note that in our case, primal feasibility is satisfied by X being rank D and w not effecting the rank of the transformed matrix since it only rescales length and not direction.

For stationarity, recall that

$$\|\beta\|_{1,2} = \sum_{p=1}^{P} \|\beta_{p.}\|_2. \tag{23}$$

Then, by the definition of subdifferential

$$\partial\|\beta\|_{1,2} = \begin{bmatrix} I_D \\ v_{d+1} \\ \dots \\ v_P \end{bmatrix} \tag{24}$$

where $v_p \in \mathbb{R}^D$ can be any vector satisfying $\|v_p\|_2 \leq 1$.

We therefore must show that there exists a ν that satisfies

$$\begin{bmatrix} I_D \\ w(X_{-S})^T \end{bmatrix} \nu = \begin{bmatrix} I_D \\ V_{-S} \end{bmatrix}. \tag{25}$$

At this point we can see that in fact I_D is an appropriate choice of ν since normalization by w leads to vectors satisfying $\|w(X)\|_2 \leq 1$.

Since for this β, $S(\beta) = S$, we have shown that S is the support of a solution to $\beta_{MBP}(w(X), I_D)$. □

The next part of the proof handles the presence of non-unique solutions.

Proposition 6 $S(\widehat{\beta}(X)) = \cup\, S(\beta) \;:\; \beta \in \beta_{MBP}(w(X), I_D)$

Proof: The proposition follows from geometric properties of the solution set. We ignore the case for singleton solutions, for which the proposition is self-evident. Recall that $\widehat{\beta}(X) := \widehat{\beta}^{\ell_2}_{MBP}(w(X), I_D) = \arg\min_{\beta \in \mathbb{R}^{P\times D}} \|\beta\|_F : \beta \in \widehat{\beta}_{MBP}(X, I_D)$. By definition of $\widehat{\beta}_{MBP}(X, I_D)$, $\|\beta_{p\cdot}\|_2 : \beta \in \widehat{\beta}_{MBP}(X, I_D)$ is an affine linear subspace of $\mathbb{R}^P$. The projection of the vector of minimum ℓ_2 distance from the origin to a affine space intersects that space perpendicularly. However, for all $\|\beta_{\cdot p}\|_2$ such that β is on the intersection of this affine space with the coordinate hyperplanes formed by fixing any choices of $\|\beta_{\cdot p}\|_2 = 0$, this vector doesn't intersect this space perpendicularly. Thus, $\widehat{\beta}(X)$ is not on the intersection of the solution polytope $\widehat{\beta}_{MBP}(w(X), I_D)$ with the coordinate hyperplanes. It is therefore maximally non-sparse with respect to any of the minimizing solutions. □

2.2.3 Proof of Proposition **??**

X_S admits a QR decomposition $X_S = QR$ where Q is orthogonal and R is upper triangular. By Proposition **??**, we can assume without loss of generality that $Q = I_D$. That is, X_S is upper triangular. Thus, $\beta = X_S^{-1}$ is upper triangular with diagonal elements $\beta_{dd} = R^{-1}_{dd}$. $\|\beta_{d\cdot}\|_2 \geq \beta_{dd}$, with equality only occurring when the off-diagonal elements are 0. Therefore, $\|\beta_{1,2}\| \geq D$ with equality occurring only when $\beta = I_D$.

2.2.4 Proof of Proposition **??**

Proof: The augmented split Lagrangian minimized by ADMM is

$$\mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) = \|z\|_{1,2} + \mathrm{tr}(\nu^T(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_F^2 + \gamma\|\beta - z\|_F^2 \tag{26}$$

This minimization problem is

$$\arg \min_{\beta, z \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z) \tag{27}$$

For a more detailed explanation of the reasons to prefer optimizing the split augmented Lagrangian over direct optimization of Equation **??** see Boyd (2010).

The three main subroutines (31 - 35) called by ADMM determine the per iteration complexity. The update rules of the ADMM algorithm used to optimize the augmented split Lagrangian are

$$\beta^{k+1} = \arg \min_{\beta \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta, z^k) \tag{28}$$

$$z^{k+1} = \arg \min_{z \in P\times D} \mathcal{L}_{\rho,\gamma,\nu}(X, \beta^{k+1}, z) \tag{29}$$

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{30}$$

We will analyze the computational complexity of each of these steps by rewriting the update rules according to the specifics of the isometry pursuit problem.

The beta update Equation 31 projects the solution onto the feasible set.

$$\beta^{k+1} = \arg\min_{\beta \in P \times D} \operatorname{tr}(\nu^{k^T}(I_D - w(X)\beta)) + \rho\|I_D - w(X)\beta\|_2^2 + \gamma\|\beta - z^k\|_2^2 \tag{31}$$

$$= (\rho\, w(X)^T w(X) + \gamma I_D)^{-1}(\rho\, w(X)^T I_D + \gamma z^k + w(X)^T \nu^k) \tag{32}$$

The computational complexity of this computation is dominated by the inverse - that is, $O(D^2P)$.

Subsequently, the z update Equation 33 is soft-thresholding operation onto a contracted level set $\|\|_{1,2}$ ball.

$$z^{k+1} = \arg\min_{z \in P \times D} \|z\|_{1,2} + \gamma\|z - \beta^{k+1}\|_2^2 \tag{33}$$

$$= \begin{cases} (1 - \frac{1}{\gamma\|\beta_{j.}^{k+1}\|_2})(\beta_{j.}^{k+1}) \text{ if } \|\beta_{j.}^{k+1}\|_2 > \frac{1}{\gamma} \\ 0 \text{ otherwise} \end{cases} \tag{34}$$

The computational complexity of this operation is $O(PD)$.

Finally, the ν update Equation 35 increases the squeezing of the z and β variables towards to each other.

$$\nu^{k+1} = \nu^k + \rho(I_D - w(X)\beta^{k+1}) \tag{35}$$

In this step, the dual variable ν that controls the penalty accrued by the splitting is updated in the direction of the constraint violation. The computational complexity of this operation is dominated by that of the matrix multiplication $w(X)\beta^{k+1}$. That is, it is $O(D^2P)$. □

2.3 Partial derivatives of candidate losses

We summarize the column-wise derivatives of the losses in Table **??**. Throughout this section let $X = [x_1, \ldots, x_D] \in \mathbb{R}^{D \times D}$ be full rank. For a scalar loss $L(X)$, write

$$\nabla_X L = \left[\frac{\partial L}{\partial x_1}, \ldots, \frac{\partial L}{\partial x_D}\right]. \tag{36}$$

All formulas are stated on smooth strata: rank deficient matrices, repeated singular values, and zero rows of pseudoinverse factors should be handled using subgradients.

We will use two common identities. First, if $X = U\operatorname{diag}(\sigma)V^\top$ is a singular value decomposition and $L(X) = \sum_{d=1}^{D} \phi(\sigma_d(X))$, then

$$\nabla_X L = U\operatorname{diag}(\phi'(\sigma_1), \ldots, \phi'(\sigma_D))V^\top. \tag{37}$$

Second, for normalized losses let $N = n_q^{\times D}(X)$. If $M = \nabla_N L$, then the column-wise chain rule is

$$\frac{\partial L}{\partial x_p} = \mathrm{D}n_q(x_p)^\top M_{.p}. \tag{38}$$

For $r = \|x\|_2$, write $n_q(x) = s(r)x$ with $s(r) = q(r)/r$. Then

$$\mathrm{D}n_q(x) = s(r)I_D + \frac{s'(r)}{r}xx^\top, \tag{39}$$

$$s'(r) = \frac{q'(r)}{r} - \frac{q(r)}{r^2}, \tag{40}$$

$$q'(r) = -\frac{2e(e^r - r^{-2}e^{1/r})}{(e^r + e^{1/r})^2}. \tag{41}$$

Since $\mathrm{D}n_q(x)$ is symmetric, the transpose in this chain rule may be omitted.

Distortion. Let $\sigma_1 = \sigma_{\max}(X)$ and $\sigma_D = \sigma_{\min}(X)$ with corresponding singular vectors u_1, v_1 and u_D, v_D. For

$$L_{\text{dist}}(X) = \frac{\sigma_1}{\sigma_D} - 1, \tag{42}$$

The spectral gradient identity gives

$$\nabla_X L_{\text{dist}} = \frac{u_1 v_1^\top}{\sigma_D} - \frac{\sigma_1}{\sigma_D^2} u_D v_D^\top. \tag{43}$$

The derivative is insensitive to uniform rescaling, which is the sense in which this loss does not prefer unit length.

Isometry loss. For

$$L_{\text{iso}}(X) = \sum_{d=1}^{D} g(\sigma_d(X)) - D, \tag{44}$$

we have

$$g'(t) = \frac{e^t - t^{-2} e^{1/t}}{2e}. \tag{45}$$

Thus

$$\nabla_X L_{\text{iso}} = U \operatorname{diag}(g'(\sigma_1), \ldots, g'(\sigma_D)) V^\top. \tag{46}$$

S-divergence to identity. Let $G = X^\top X$ and

$$L_{\text{sdiv}}(X) = \log\det\left(\frac{G + I_D}{2}\right) - \frac{1}{2}\log\det(G). \tag{47}$$

The constant $\log\det(1/2)$ does not contribute to the derivative. Using $d\log\det A = \operatorname{tr}(A^{-1}dA)$ and $dG = dX^\top X + X^\top dX$,

$$\nabla_X L_{\text{sdiv}} = 2X(G + I_D)^{-1} - XG^{-1}. \tag{48}$$

Normalized log-det volume. Let $N = n_q^{\times D}(X)$ and

$$L_{\text{nld}}(X) = -\sum_{d=1}^{D} \log \sigma_d(N). \tag{49}$$

For square full-rank N, this is $-\log|\det N|$, so

$$\nabla_N L_{\text{nld}} = -N^{-\top}. \tag{50}$$

Therefore, by the column-wise chain rule,

$$\frac{\partial L_{\text{nld}}}{\partial x_p} = -\mathrm{D}n_q(x_p)^\top (N^{-\top})_{\cdot p}. \tag{51}$$

Frobenius pseudoinverse. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. For

$$L_{\text{frob}}(X) = \|N^{-1}\|_F - \sqrt{D}, \tag{52}$$

the differential identity $dB = -B(dN)B$ gives

$$\nabla_N L_{\text{frob}} = -\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F}. \tag{53}$$

Thus

$$\frac{\partial L_{\text{frob}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top \left(-\frac{N^{-\top} N^{-1} N^{-\top}}{\|N^{-1}\|_F}\right)_{\cdot p}. \tag{54}$$

Isometry pursuit proxy. Let $N = n_q^{\times D}(X)$ and $B = N^{-1}$. The differentiable proxy for the IsometryPursuit table row is

$$L_{\mathrm{ip}}(X) = \|N^{-1}\|_{1,2} - D. \tag{55}$$

Define $R \in \mathbb{R}^{D\times D}$ by

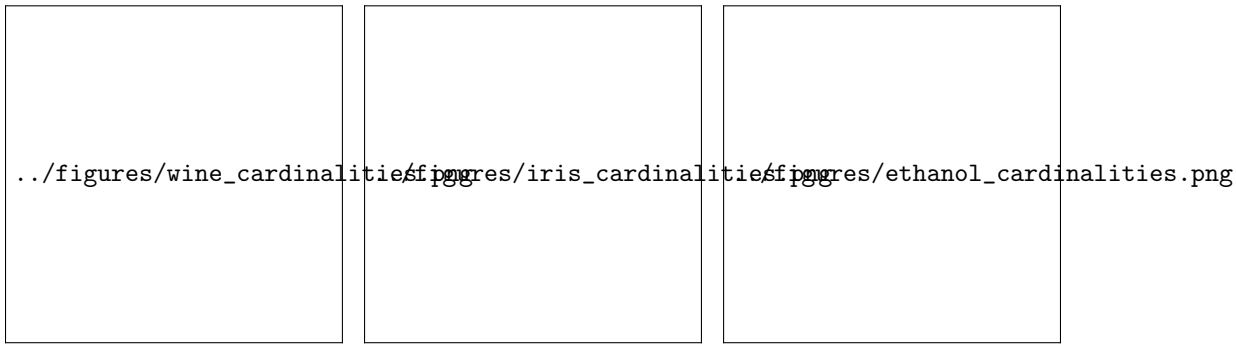
$$R_{p\cdot} = \frac{B_{p\cdot}}{\|B_{p\cdot}\|_2}. \tag{56}$$

Then $R \in \partial\|B\|_{1,2}$ on the smooth stratum where every row of B is nonzero, and

$$\nabla_N L_{\mathrm{ip}} = -B^\top R B^\top. \tag{57}$$

The column-wise gradient is therefore

$$\frac{\partial L_{\mathrm{ip}}}{\partial x_p} = \mathrm{D}n_q(x_p)^\top (-B^\top R B^\top)_{\cdot p}. \tag{58}$$



(a) Wine (b) Iris (c) Ethanol

Figure 1: Support cardinalities for each dataset after IsometryPursuit.

2.4 Support cardinalities

Figure 1 plots the distribution of $|\widehat{S}_{IP}|$ from Table **??** in order to contextualize the reported means. While typically $|\widehat{S}_{IP}| << P$, there are cases for Ethanol where this is not the case that drive up the means. Further investigation into the extreme failure of isometry pursuit to sparsify in the the minor chunk of the bimodal Ethanol support cardinality distribution is warranted.

2.5 Timing

While wall-time of algorithms is a non-theoretical quantity that depends on implementation details, it provides valuable context for practitioners. We therefore report the following runtimes on a 2021 Macbook Pro. The particularly high variance for brute force search in the second step of TwoStageIsometryPursuit is likely due to the large cardinalities reported in Figure 1.

Name	IP	2nd stage	Greedy
Iris	1.24 ± 0.02	0.00 ± 0.00	0.02 ± 0.00
Wine	2.32 ± 0.17	0.13 ± 0.12	0.03 ± 0.00
Ethanol	8.38 ± 0.57	0.55 ± 1.08	0.07 ± 0.01

Table 1: Algorithm runtimes in seconds across replicates.

Example 1

$$X_S = \begin{bmatrix} \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} & \frac{1}{2} & \frac{-1}{2} \\ \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 & & \\ 0 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & 0 & 0 \\ 0 & 00 & 0 & \frac{\sqrt{3}}{2} & \frac{\sqrt{3}}{2} & \end{bmatrix}. \tag{59}$$

Proof: Let

By Given the symmetry of the second and third columns around the y axis, the loss will not change regardless of whether we upweight one or downweight the other, these changes necessarily being in conjunction. We therefore assume that their loadings with respect to the second column of the target matrix (the D dimensional identity) are the same. Clearly, they will both then have value of $\frac{1}{\sqrt{3}}$. The question then becomes, what should the values of the first column in the loading matrix β be?

We can answer this question with a simple differential equation. Let b denote the constraint violation $1 - X_{1.}\beta_{.1}$.

Then

$$\frac{\partial b}{\partial \beta_{11}} = \frac{\partial b}{\partial \beta_{21}} \tag{60}$$

$$= \frac{1}{2} \tag{61}$$

On the other hand, due to the value of $\frac{1}{\sqrt{3}}$ already assumed for the second loading of the second and third features,

$$\frac{\partial l}{\partial \beta_{11}} = \frac{\partial l}{\partial \beta_{21}} \tag{62}$$

$$= \frac{\partial}{\partial x}\sqrt{\frac{1}{3} + x^2} \tag{63}$$

$$= \frac{1}{2}(\frac{1}{3} + x^2)^{-\frac{1}{2}}(2x) \tag{64}$$

$$= \frac{x}{\sqrt{\frac{1}{3} + x^2}}. \tag{65}$$

This means that the tradeoff

$$\frac{\partial l}{\partial b} = \frac{1}{2x}\sqrt{\frac{1}{3} + x^2}. \tag{66}$$

In contrast, $\frac{\partial l}{\partial \beta_{11}} = 1$ and similarly for its constraint violation. Solving for the equilibrium, we get

$$\frac{1}{2x}\sqrt{\frac{1}{3} + x^2} = 1 \tag{67}$$

$$\frac{1}{3} + x^2 = 4x^2 \tag{68}$$

$$\frac{1}{3} = 3x^2 \tag{69}$$

$$x = \frac{1}{\sqrt{3}}. \tag{70}$$

This leads to a value for l of

$$\sqrt{\frac{1}{3} + \frac{1}{9}} + \sqrt{\frac{1}{3} + \frac{1}{9}} + \frac{2}{3} \tag{71}$$

$$= \sqrt{\frac{4}{9}} + \sqrt{\frac{4}{9}} + \frac{2}{3} \tag{72}$$

$$= \frac{2}{3} + \frac{2}{3} + \frac{2}{3} \tag{73}$$

□